MATH 3312, SPRING 2026: SOLUTIONS TO HOMEWORK 9

Definition 1. If R is a commutative ring, M is an R-module and $a \in R$, then we set
$$a \cdot M = aM = \{a \cdot m : \ m \in M\}.$$
This is an R-submodule of M (why?).

(1) Let R be a PID. Then, for $a, b \in R$, show that the map $R \xrightarrow{r \mapsto ar+(b)} R/(b)$ yields an isomorphism
$$R/(c) \xrightarrow{\simeq} a \cdot (R/(b)),$$
where $c = b/\gcd(a,b)$.[1]

Solution. Let's call the given map $f : R \to R/(b)$. The image of f is exactly $a \cdot (R/(b))$, and so the first isomorphism theorem gives us an isomorphism
$$R/\ker(f) \xrightarrow{\simeq} a \cdot (R/(b)).$$
To figure out what the kernel is, note that $ar + (b) = (b)$ is zero if and only if $ar \in (b)$. Certainly if r is a multiple of $c = b/\gcd(a,b)$, then $ac = \frac{a}{\gcd(a,b)} b$ is a multiple of b, and so $ar \in (b)$.

Conversely, if $ar = bs$, then dividing both sides by $d = \gcd(a,b)$, tells us that $\frac{a}{d} r = \frac{b}{d} s = cs$.[2] But now $\frac{a}{d}$ and $c = \frac{b}{d}$ have no common prime factors, and so the only way this equality can hold is if all the prime factors of $\frac{b}{d}$ show up in r. In other words, $c \mid r$.

This shows that $\ker(f) = (c)$, as desired.

Definition 2. Let R be a commutative ring and let M be an R-module. A **chain of submodules of length** n of an R-module M is a sequence
$$M_n < M_{n-1} < \cdots < M_1$$
of *non-zero* submodules of M where the $<$ denotes *strict* inclusion (that is, M_i is a *proper* submodule of M_{i-1}).

M has **length** n for $n \geq 1$ if there is a chain of submodules of length n, but no chain of length $n+1$. In this case, we will set $\mathrm{length}(M) = n$. If $M = \{0\}$, we will set $\mathrm{length}(M) = 0$. In all these cases, we will say that M has **finite length**.

(2) Suppose that M is an R-module of finite length. If $N \leq M$ is an R-submodule, show that N and M/N also have finite length, and that in fact, we have
$$\mathrm{length}(N) + \mathrm{length}(M/N) = \mathrm{length}(M).$$

Solution. We will prove two inequalities:

(a) $\mathrm{length}(N) + \mathrm{length}(M/N) \leq \mathrm{length}(M)$.

For this, suppose that we have a chain
$$N_r < N_{r-1} < \cdots < N_1$$
of submodules of N and a chain
$$\overline{M}_s < \overline{M}_{s-1} < \cdots < \overline{M}_1$$

[1] Note that the gcd is only well-defined up to invertible elements.

[2] Here, and elsewhere, when we write $\frac{a}{d}$, what we mean is the unique element such that $d \cdot \frac{a}{d} = a$, which exists since a is a multiple of d by hypothesis.

of M/N. Then there exists a chain $M_s < M_{s-1} < \cdots < M_1$ of submodules of M containing N such that $\overline{M}_j = M_s/N$. We can combine the first chain with this one to get a chain of length $r+s$ in M. This gives us the first inequality.

(b) $\mathrm{length}(N) + \mathrm{length}(M/N) \geq \mathrm{length}(M)$
Suppose that we have some chain

$$M_1 < \cdots < M_n$$

We can intersect this chain with N and throw away any repeated submodules and get a chain

$$N_1 < \cdots < N_r.$$

We can also take the images $\pi(M_i) \leq M/N$ where $\pi : M \to M/N$ is the quotient homomorphism, throw away the repeated suhmodules and get a chain

$$\overline{M}_1 < \cdots < \overline{M}_s$$

of submodules of M/N.
We will need the following observation:

Observation 1. If $M_1 \leq M_2$ are submodules of M such that $M_1 \cap N = M_2 \cap N$ and $\pi(M_1) = \pi(M_2) \leq M/N$. Then $M_1 = M_2$.

Proof. We want to show that every element $m \in M_2$ is already in M_1. Since $\pi(M_1) = \pi(M_2)$, we know that the coset $\pi(m) = m + N$ is of the form $\pi(m') = m' + N$ with $m' \in M_1$. This means that $m - m' \in N$ (why?) and so we have $m = m' + n$ where $m' \in M_1$ and $n \in N$. But now $n = m - m' \in M_2 \cap N = M_1 \cap N \leq M_1$. But then we have written m as the sum of two elements in M_1, and so $m \in M_1$, as desired. □

Then the observation tells us that, since $M_i < M_{i+1}$ for every $i < n$, we must either have $M_i \cap N \neq M_{i+1} + N$ *or* $\pi(M_i) \neq \pi(M_{i+1})$. In other words, for every successive pair in the chain, at worst we throw away *one of* the intersection with N and the image under π from the pair. This implies that we must have $r + s = n$. This gives us the second inequality and finishes the proof.

(3) If R is a PID and $q \in R$ is an irreducible element, show that $R/(q^m)$ has length m as an R-module, and conclude that $q^r \cdot (R/q^m)$ has length $\max(0, m-r)$.

Solution. We will prove this by induction on $m \geq 1$. If $m = 1$, then $R/(q)$ is a field and so has no non-zero proper submodules because $(q) \leq R$ is maximal. If $m \geq 2$, then we have a submodule $(q)/(q^m) \leq R/(q^m)$ where the quotient is $R/(q)$. Therefore, by the previous problem, we have

$$\mathrm{length}(R/(q^m)) = \mathrm{length}((q)/(q^m)) + \mathrm{length}(R/(q)).$$

But $(q)/(q^m) = q\cdot(R/(q^m)) \simeq R/(q^{m-1})$ by problem 1. Therefore, by induction, we have $\mathrm{length}((q)/(q^m)) = \mathrm{length}(R/(q^{m-1})) = m-1$. This tells us

$$\mathrm{length}(R/(q^m)) = \mathrm{length}((q)/(q^m)) + \mathrm{length}(R/(q)) = (m-1) + 1 = m.$$

Now, by problem 1, $q^r \cdot R/(q^m) = R/(q^m/\gcd(q^m, q^r))$. We have

$$\gcd(q^m, q^r)) = q^{\min(m,r)},$$

and so $q^m/\gcd(q^m, q^r)) = \max(1, q^{m-r})$. This tells us that $q^r \cdot R/(q^m)$ has length 0 if $r \geq m$ and has length $m - r$ otherwise.

(4) Suppose that $p(x) \in \mathbb{Z}[x]$ is a *non-constant* irreducible element, and that $h_1(x), h_2(x) \in \mathbb{Z}[x]$ are such that $h_1(x)h_2(x) \in (p(x))$.

(a) Show that $p(x)$ is primitive (see Problem 7 on Homework 8) and that it is also irreducible as an element of $\mathbb{Q}[x]$.

Solution. If $p(x)$ is not primitive, then we would have $p(x) = d \cdot p_1(x)$ where $d > 1$ is the gcd of the coefficients. But since d is non-invertible and $p_1(x)$ is non-constant and hence also non-invertible, this would contradict the irreducibility of $p(x)$.

If $p(x)$ factors as $p(x) = f(x)g(x) \in \mathbb{Q}[x]$ with $f(x)$ and $g(x)$ non-constant, then as in Homework 8 we can find $r_1, r_2 \in \mathbb{Q}_{>0}$ such that $r_1 f(x)$ and $r_2 g(x)$ are primitive non-constant polynomials in $\mathbb{Z}[x]$. But then $r_1 r_2 p(x) = (r_1 f(x))(r_2 g(x)) \in \mathbb{Z}[x]$. Moreover, as the product of primitive polynomials $r_1 r_2 p(x)$ is also primitive.

Observation 2. If $p(x)$ is a primitive polynomial and $r \in \mathbb{Q}_{>0}$ is such that $rp(x)$ is also primitive, then we have $r = 1$.

Proof. If $r = \frac{a}{b}$ with $a, b \in \mathbb{Z}_{\geq 1}$, then the gcd of the coefficients of $b(rp(x)) = ap(x)$ is a (because the gcd of the coefficients of $p(x)$ is 1), but it is also b (because the gcd of the coefficients of $rp(x)$ is 1). This tells us that $a = b$ and hence $r = 1$. $\square$

The observation tells us that $r_1 r_2 = 1$, and so we have $p(x) = (r_1 f(x))(r_2 g(x))$ contradicting the irreducibility of $p(x)$.

(b) Show that, after switching the indices if necessary, there exists $g(x) \in \mathbb{Q}[x]$ such that $h_1(x) = p(x)g(x)$.

Solution. Because $p(x)$ is irreducible in $\mathbb{Q}[x]$, it is prime in $\mathbb{Q}[x]$. Since $h_1(x)h_2(x)$ is a multiple of $p(x)$ in $\mathbb{Z}[x]$, it is also a multiple in $\mathbb{Q}[x]$. The prime-ness of $p(x)$ in $\mathbb{Q}[x]$ now tells us that either $h_1(x)$ or $h_2(x)$ is a multiple of $p(x)$ in $\mathbb{Q}[x]$. By switching indices, we can assume that

$$h_1(x) = p(x)g(x)$$

for $g(x) \in \mathbb{Q}[x]$.

(c) Show that $g(x) \in \mathbb{Z}[x]$ and therefore that $p(x)$ is a *prime* element of $\mathbb{Z}[x]$.

Solution. Choose $r \in \mathbb{Q}_{\geq 0}$ such that $rg(x) \in \mathbb{Z}[x]$ is primitive. Then

$$rh_1(x) = p(x)(rg(x)) \in \mathbb{Z}[x]$$

is primitive, since it is the product of two primitive polynomials. If $r = \frac{a}{b}$ in least terms, then this tells us that

$$ah_1(x) = p(x)(ag(x)) \in \mathbb{Z}[x].$$

The gcd of the coefficients of $ag(x)$ is b (since $\frac{a}{b}g(x)$ is primitive). Since $p(x)$ is primitive, this implies that the gcd of the coefficients of $ah_1(x)$ is also b. Since a and b are relatively prime, and a clearly divides all the coefficients of $h_1(x)$, this can only happen if $a = 1$. Therefore, $1 \cdot g(x)$ is in $\mathbb{Z}[x]$, as desired.

(d) Conclude that $\mathbb{Z}[x]$ is a UFD.

Solution. By the Observation in Lecture 22, this amounts to knowing that every irreducible element of $\mathbb{Z}[x]$ is prime. What we have seen above shows that every non-constant irreducible element of $\mathbb{Z}[x]$ is prime. A *constant* irreducible element of $\mathbb{Z}[x]$ is just an irreducible element of $\mathbb{Z}$ and so is prime since $\mathbb{Z}$ is a UFD.

Hint: The argument for part (a) is like that used for Gauss's lemma in Homework 8.

Definition 3. Suppose that k is a field and that $A \in M_{n\times n}(k)$. For each $\lambda \in k$, the **algebraic multiplicity** of λ is the largest non-negative integer $m_A^{\mathrm{alg}}(\lambda)$ such that $(x-\lambda)^{m_A^{\mathrm{alg}}(\lambda)}$ divides $\mathrm{ch}_A(x)$.

The **geometric multiplcity** of λ is the quantity

$$m_A^{\mathrm{geom}}(\lambda) = \dim_k(\ker(A - \lambda I_n)).$$

(5) Suppose that $A \in M_{n\times n}(k)$ and that $\lambda \in k$. Show that we always have

$$m_A^{\mathrm{geom}}(\lambda) \leq m_A^{\mathrm{alg}}(\lambda)$$

and that the following are equivalent:

(a) $m_A^{\mathrm{alg}}(\lambda)$ is non-zero.
(b) $m_A^{\mathrm{geom}}(\lambda)$ is non-zero.

(c) λ is an eigenvalue.

Hint: For the first part, it might be useful to look at the elementary divisor decomposition.

Solution. Suppose that V_A is has the elementary divisor decomposition

$$V_A \simeq \prod_{i=1}^{r} k[x]/(q_i(x)^{r_i}).$$

Then the action of $A - \lambda I_n$ on the left hand side translates to multiplication by $x - \lambda$ on the right. As seen in Problem 4 of Homework 8, $x - \lambda$ acts injectively on $k[x]/(q_i(x)^{r_i})$ unless $q_i(\lambda) = 0$. But since $q_i(x)$ is *irreducible*, this lack of injectivity is only possible if $q_i(x) = x - \lambda$.

Therefore, $\ker(x - \lambda) \neq 0$ if and only if there exists i such that $q_i(x) = x - \lambda$. Moreover, in this case, the kernel of $x - \lambda : k[x]/((x - \lambda)^{r_i}) \to k[x]/((x - \lambda)^{r_i})$ is exactly 1-dimensional (why?).

Thus we see that $m_A^{\mathrm{geom}}(\lambda) = \dim_k \ker(x - \lambda)$ is exactly the number of indices i such that $q_i(x) = x - \lambda$. On the other hand, the algebraic multiplicity is the sum of the r_i over such indices i. This tells us $m_A^{\mathrm{geom}}(\lambda) \leq m_A^{\mathrm{alg}}(\lambda)$.

The equivalence of the three statements is now just a reinterpretation of the fact that λ is an eigenvalue if and only if $A - \lambda I_n$ is not injective, which holds if and only if $x - \lambda$ acts non-injectively on the elementary divisor decomposition.

Definition 4. A matrix $B \in M_{n\times n}(k)$ is in **Jordan canonical form** if there exist $\lambda_1, \dots, \lambda_r \in k$ and integers $m_1, \dots, m_r \in \mathbb{Z}_{\geq 1}$ such that A is a block diagonal matrix of the form

$$B = \begin{bmatrix} J_{\lambda_1}(m_1) & 0 & \cdots & 0 \\ 0 & J_{\lambda_2}(m_2) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & J_{\lambda_r}(m_r) \end{bmatrix}$$

Here, the diagonal entries are the Jordan block matrices from Homework 7.

(6) Show that the following are equivalent:
  (a) A is k-conjugate to a matrix in Jordan canonical form.
  (b) $\mathrm{ch}_A(x)$ splits into a product of linear factors in $k[x]$.
  (c) $\min_A(x)$ splits into a product of linear factors in $k[x]$.

Solution. Saying that $\mathrm{ch}_A(x)$ or $\min_A(x)$ splits into a product of linear factors is equivalent to saying that all the elementary divisors $q_i(x)^{r_i}$ are of the form $(x - \lambda_i)^{r_i}$ with λ_i an eigenvalue (why is this?).

On the other hand, by Problem 5 on Homework 7 and Problem 5 on Homework 6, saying that A is k-conjugate to a matrix in Jordan canonical form is equivalent to saying that V_A is isomorphic to a product of the form $\prod_i k[x]/((x - \lambda_i)^{m_i})$. This is of course equivalent to saying that all the elementary divisors are linear.

(7) Show that the following are equivalent:
  (a) A is diagonalizable.
  (b) $\min_A(x)$ splits into a product of *distinct* linear factors in $k[x]$.

Hint: What does the minimal polynomial have to do with the Jordan canonical form? It's best to think module theoretically for such things.

Solution. First, note:

Observation 3. The minimal polynomial is the product of the *highest* powers of the *distinct* irreducible polynomials showing up in the elementary divisor decomposition.

If A is diagonalizable, then by Problem 4 on Homework 7, all its elementary divisors are linear of the form $x - \lambda_i$. Therefore, the minimal polynomial is just a product of the *distinct* linear terms.

Conversely, if the minimal polynomial is a product of distinct linear terms, then the elementary divisor decomposition only contains linear terms, and the corresponding Jordan canonical form (which exists by the previous problem) will only have 1×1 blocks and will therefore just be a diagonalizable matrix.

(8) Suppose that L/k is a field extension and that $f(x), g(x) \in k[x]$ and $h(x) \in L[x]$ are non-zero polynomials such that $f(x) = h(x)g(x)$. Show that $h(x) \in k[x]$.

Solution. By the division algorithm, we have $f(x) = q(x)g(x) + r(x)$ with $q(x), r(x) \in k[x]$ and either $r(x) = 0$ or $\deg r(x) < \deg g(x)$. Equating

$$h(x)g(x) = f(x) = q(x)g(x) + r(x) \in L[x]$$

gives us $r(x) = (h(x) - q(x))g(x) \in L[x]$, which by degree considerations tells us that $r(x) = 0$ and hence $h(x) - q(x) = 0$, implying that $h(x) = q(x) \in k[x]$.

(9) Prove *Eisenstein's criterion*: If R is a PID, $q \in R$ is irreducible, and $f(x) = x^d + a_{d-1}x^{d-1} + a_{d-2}x^{d-2} + \cdots + a_1x + a_0 \in R[x]$ is such that $q \mid a_i$ for all $i = 0, \ldots, d-2$ but $q^2 \nmid a_0$, then $f(x)$ is irreducible. Use this to verify that the following polynomials in $\mathbb{Q}[x]$ are irreducible:
(a) $x^3 - 20x + 15$.
(b) $x^3 + 3x^2 + 3x - 1$.
(c) $x^2 - x - 1$.
(d) $x^6 + x^5 + x^4 + x^3 + x^2 + x + 1$.
Hint: If $f(x)$ admits a factorization, what can you say about the factors after quotienting R by (q)? Also, for a couple of the examples, you might want to play around with substitutions for x.

Solution. For Eisenstein's criterion, suppose that we had a factoring $f(x) = g(x)h(x) \in R[x]$. We can assume that $g(x)$ and $h(x)$ both have leading coefficient 1 (why?).

Write $\overline{f(x)}, \overline{g(x)}, \overline{h(x)}$ for the images of these polynomials in $(R/(q))[x]$ The conditions on the coefficients of $f(x)$ tell us $\overline{f(x)} = x^d$. But then, since $R/(q)$ is a field, unique factorization tells us that we must have $\overline{g(x)} = x^m$ and $\overline{h(x)} = x^{d-m}$ for some $m \le d$. This tells us that q divides all the lower degree coefficients of both $g(x)$ and $h(x)$. In particular, it divides the constant terms $g(0)$ and $h(0)$. But now the constant term of $f(x)$ is $a_0 = f(0) = g(0)h(0)$. If $g(x)$ and $h(x)$ are both non-constant, then this would tell us that q^2 divides a_0, which is a contradiction.

For the examples, first note that, Gauss's lemma, an integer polynomial with leading coefficient 1 is irreducible in $\mathbb{Q}[x]$ if and only if it is irreducible in $\mathbb{Z}[x]$. Therefore, we can apply Eisenstein's criterion.

For (a), you can apply it directly with $p = 5$.

For (b), note that you can write this as

$$x^3 + 3x^2 + 3x - 1 = x^3 + 3x^2 + 3x + 1 - 2 = (x+1)^3 - 2.$$

Set $y = x + 1$ and apply Eisenstein to $y^3 - 2$ to see that it is irreducible as a polynomial in y, and hence that the original polynomial is irreducible as a polynomial in x.

(c) can be deduced directly by noting that it is a quadratic with no rational zeros (use the quadratic formula). Alternately, note that if $y = x + 1$, then we can rewrite the polynomial as

$$(y-1)^2 - (y-1) - 1 = y^2 - 2y + 1 - y - 1 - 1 = y^2 - 3y - 3,$$

to which we can apply the criterion with $p = 3$.

For (d), note that

$$x^6 + x^5 + x^4 + x^3 + x^2 + x + 1 = \frac{x^7 - 1}{x - 1}.$$

Set $y = x - 1$, and rewrite the right hand side as

$$\frac{(y+1)^7 - 1}{y} = \frac{\sum_{i=0}^{6} \binom{7}{i} y^{7-i}}{y} = \sum_{i=0}^{6} \binom{7}{i} y^{7-i-1}$$

This polynomial satisfies criterion with $p = 7$.

Definition 5. A polynomial $f(x) \in k[x]$ is **separable** if $\gcd(f(x), f'(x)) = 1$ (see Proposition in Lecture 9).

(10) Let $f(x) \in k[x]$ is an irreducible and separable polynomial of degree d. Show that the following are equivalent for an extension $k \leq L$:
  (a) $f(x)$ splits into a product of linear factors in $L[x]$.
  (b) $|\mathrm{Emb}_k(k[x]/(f(x)), L)| = d$.

  See Lecture 5 for the definition of Emb_k.

Solution. Since

$$\mathrm{Emb}_k(k[x]/(f(x)), L) \simeq \{\alpha \in L : \ f(\alpha) = 0\},$$

the second statement is equivalent to saying that $f(x)$ has exactly d distinct zeros in L.

On the other hand, the separability condition implies that $f(x)$ does not have repeated zeros in L (this is the Proposition in Lecture 9), and so if it splits into a product of linear factors in $L[x]$, it actually splits into a product of *distinct* linear factors in $L[x]$. In other words, the first statement is also equivalent to $f(x)$ having d distinct zeros in L.